Harshwardhan Patil

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Overview

Full-Stack Engineer with **3+ years of experience** building scalable, distributed systems and high-performance APIs across frontend, backend, and cloud environments. Skilled in **system design**, CI/CD on AWS, production monitoring, and delivering **fault-tolerant, data-consistent services** at scale.

Technical Skills

- **Languages & Frameworks:** Java, TypeScript, JavaScript, SQL, Spring Boot, Angular, Node.js
- **Cloud & DevOps:** AWS (EC2, ECS, S3, RDS, CloudFront, ALB, Secrets Manager, Lambda), Docker, Jenkins, GitHub Actions, Nginx, Cloudflare, CI/CD
- **Databases:** PostgreSQL, Redis, MySQL, Neo4j, Elasticsearch, MongoDB, Schema design, Database indexing, Query optimization, Caching strategies
- **Testing & Security:** JUnit, Mockito, Jasmine, SonarQube, Postman, JWT, RBAC, HTTPS/TLS, CORS, CSRF/XSS protection
- **Design & Architecture:** System Design, Design Patterns, SOLID principles, Distributed backend services, Concurrency handling, Session management

Professional Experience

Software Engineer, Clarivate Analytics – Bangalore, India *Aug 2023 – Jul 2024*

- Developed scalable Spring Boot REST APIs with DTO contracts and standardized error handling, reducing breaking changes by **30%** across releases.
- Implemented **JWT**-based multi-session management across backend services, capping concurrent logins at 12 per user with token invalidation on breach.
- Optimized **PostgreSQL** schemas with composite indexing across reporting and search workloads for drug and patent datasets, cutting full scans.
- Containerized and deployed microservices to **EC2** using Docker and Jenkins **CI/CD** pipelines, supporting **50+ production releases** with **zero rollbacks**.
- Enforced role-based access control across backend and frontend layers, applying tiered and region-based data visibility for **10+ access variants**.
- Implemented low-latency **Elasticsearch**-powered typeahead and autocomplete, improving search response times by **40%** over large-scale datasets.
- Developed dynamic PPT and PDF export pipelines using **PptxGenJS** and **jsPDF**, generating **100+ customized reports** with user-selected filters.
- Optimized **PostgreSQL** query logic for PPT export, decoupling bulk multi-filter chart fetching from single-chart UI rendering to reduce DB round-trips.
- Achieved **90%** test coverage using **JUnit** and **Mockito** for backend and **Jasmine** for Angular, covering unit, branch, and line coverage across all modules.

Associate Software Engineer, Clarivate Analytics – Bangalore, India *Aug 2021 – Jul 2023*

- Integrated **Elasticsearch** into Spring Boot, enabling advanced full-text queries across **millions of drug records** and reducing search latency by **30%**.
- Migrated **5+ core business flows** from Angular 8 to **Angular 13**, rebuilding components with a modular architecture and improved state management.
- Built secure request-handling across backend services using **Spring Security**, **CORS/CSRF**, and input validation to guard against unauthorized access.
- Improved responsiveness of data-heavy chart and table views by **20–25%** through lazy loading, **RxJS** streams, and Angular lifecycle optimizations.
- Reduced page load times by **20%** by applying component-level **OnPush** change detection and optimizing data-fetching patterns across data-heavy views.
- Built **D3.js** chart components with grouping and filtering aggregations for **Angular**, powering **15+ analytics views** across drug intelligence dashboards.
- Architected a configurable **Angular** table framework with sorting, pagination, and filters; adopted across **3 internal teams**, cutting duplication by **50%**.

Education

University of Texas at Arlington *Master's in Software Engineering* *GPA: 3.6 / 4.0* *May 2026*

Projects

TinyURL — URL Shortener *Feb 2026 – Mar 2026*

Production-grade URL shortening service with QR code generation, link expiry, and a fully automated cloud deployment pipeline.

Tools: Spring Boot, Angular, PostgreSQL, AWS (EC2, RDS, S3, CloudFront, ALB), Docker, Cloudflare, Lambda, EventBridge, SNS, SQS DLQ, CloudWatch

- Deployed **distributed architecture** separating SPA (S3 + CloudFront) from REST API (EC2 + ALB) with **Cloudflare** for DNS, SSL, and DDoS protection.
- Hardened the API layer with **Nginx-level rate limiting**, applying per-IP controls on URL creation, redirects, and active connections to reduce abuse risk.
- Blocked unsafe traffic before application execution using request-size constraints, security headers (**HSTS, X-Frame-Options, CSP**), and scanner filters.
- Designed serverless infrastructure scheduler — reducing monthly EC2/RDS costs by 40% with full observability and zero operational overhead.
- Built **REST APIs** for URL shortening with configurable **TTL-based expiry**, short-code generation, and redirect resolution, backed by **PostgreSQL on RDS**.
- Implemented a **CI/CD pipeline** via GitHub Actions to automate production builds, S3 sync, and **CloudFront cache invalidation** on every push to main.

Employee Management System *Dec 2024 – May 2025*

Enterprise-grade system for managing employees, teams, projects, and access control with production-grade deployment and observability.

Tools: Spring Boot, Angular 19, PostgreSQL, Redis, AWS EC2, SSM, Docker, Docker Compose, Nginx, GitHub Actions

- Engineered a multi-tenant **JWT-based RBAC** backend across 5 entity types with **AWS Secrets Manager**; deployed on **EC2** with all services containerized.
- Implemented **REST APIs** with pagination, filtering, and search; built a reusable **Angular** table component with sorting and pagination powering list views.

RingNet — Graph-Based Fraud Ring Detection *Apr 2026 - Present*

Property graph system detecting fraud networks through shared identifiers and multi-hop traversal — where graph databases outperform relational.

Tools: Neo4j, Java, Cypher, Docker, Maven, Graph Data Science Library

- Modeled fraud rings as a **Neo4j property graph**; 6-hop **Cypher** traversal detects connected networks at constant complexity vs exponential recursive SQL.
- Implemented composite **risk scoring** per account using **GDS**, weighting hop distance from fraud nodes, identifier overlap, and transaction velocity.

Battle Arena – Real-Time Multiplayer Platform *May 2025 – Present*

Real-time multiplayer system featuring a modular game engine, physics-based simulation, and distributed backend services.

Tools: Angular, Phaser 3, Spring Boot, Node.js, MongoDB, Socket.io, Docker, AWS

- Architected a **5-service microservices** backend with **Socket.io** WebSockets for real-time sync, **Redis** matchmaking queues, and **Nginx** API gateway.
- Designed a modular game engine with **Matter.js** physics, **Phaser 3** rendering, and real-time bi-directional player position sync across active sessions.